Seat No: _____________ **Enrollment No:** ___________________

PARUL UNIVERSITY

FACULTY OF ENGINEERING & TECHNOLOGY

B.Tech. Winter 2018 - 19 Examination

Semester: 5 **Date: 29/11/2018**
Subject Code: 03105303 **Time: 10:30 am to 1:00 pm**
Subject Name: Theory of Computation **Total Marks: 60**

Instructions:
1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

Q.1 Objective Type Questions - (All are compulsory) (Each of one mark) **(15)**

1. The total number of One length substrings possible for a string 'VAIBHAVI' are ___________________

2. Let n is the length of the string taken by mealy machine then which of the following shows the correct ratio between length of input taken and length of output generated by mealy machine

(a) 1 (b) n:n+1 (c) n+1:n (d) None of these

3. The number of languages accepted by DFA and NFA is same. **(TRUE/FALSE).**

4. From the options given below, the pair having different expressive power is

(a) Deterministic Push down Automata (DPDA) and Non-deterministic Push Down Automata (NPDA).
(b) Deterministic Finite Automata (DFA) and Non-deterministic Finite Automata(NFA)
(c) Deterministic turning machine and Non-Deterministic turning machine.
(d) All of the above

5.The language described by the regular expression (0+1)*0(0+1)*0(0+1)* over the alphabet {0 1} is the set of

(a) All strings containing at least two 1's
(b) All strings containing at least two 0's
(c) All strings that begin and end with either 0's or 1's
(d) All strings containing the substring 00

6. Each finite language is regular language **(TRUE/FALSE).**

7. Regular languages are closed under
(a) Union , Intersection ,Concatenation Operation
(b) Union ,Intersection, Subset operation
(c) Kleen closure, positive closure, subset operation
(d) None of the above

8. Which of the following statement is false?
(a) Context free language is the subset of context sensitive language
(b) Regular language is the subset of context sensitive language
(c) Recursively enumerable language is the super set of regular language
(d) Context sensitive language is a subset of context free language

9. A→aB ,B→bC ,C→c is the _________ grammar.

10. The deterministic finite automata has exactly one path to accept a string **(TRUE/FALSE).**

11. There exists an algorithm to determine whether two context free grammars generate the same Language. **(TRUE/FALSE).**

12.L={$a^n b^n$ | n>=0} is not a ______________Language.

13. Context free languages(CFG) are closed under Union operation but recursive language are not closed under kleen closure operation. **(TRUE/FALSE).**

14. Write name of two decision Properties of CFL.

15. Write name of any two closure properties for recursive language.

Q.2 Answer the following questions. (Attempt any three) **(15)**

A) Prove that following CFG is Ambiguous and convert it into unambiguous.

$S \rightarrow S + S \mid S * S \mid (S) \mid a$

B) Give transition table and transition diagram for deterministic PDA recognizing the following Language L.

$L=\{ a^n b^{n+m} a^m \mid n,m>=0\}$

C) Using Principle of Mathematical Induction, prove that for every n >= 1,

$$\sum_{i=0}^{n} i = n (n+1) / 2$$

D) Draw Minimum Finite Automata (MFA) for following languages:

L1 = {x / 00 is not a substring of x }

L2 = {x / x ends with 01 }

Q.3 A) Check whether the given grammar is in CNF **(07)**

S-> bA|aB

A-> bAA|aS|a

B-> aBB|bS|b

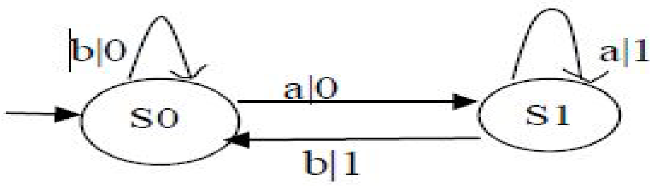
If it is not in CNF, Find the equivalent CNF.

B) Draw the TM which recognize words of the form $\{ a^n b^n c^n \mid n>=1\}$. **(08)**

OR

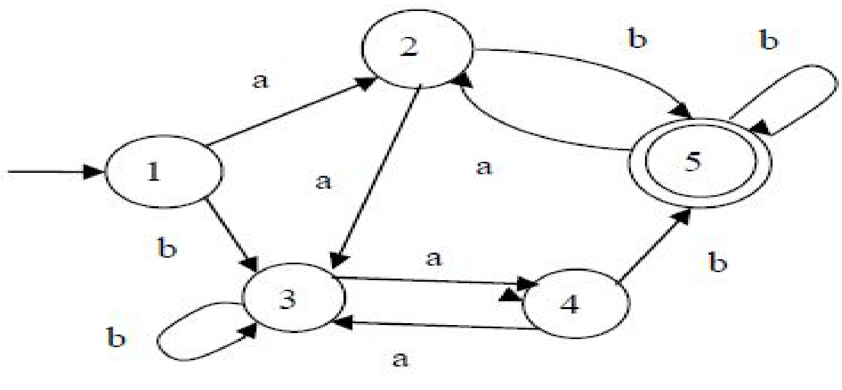
B) Define Push Down Automata (PDA). Design and draw a PDA (Push Down Automata) accepting Strings with more a's than b's. Trace it for the string "abbabaa". **(08)**

Q.4 A) Convert the Mealy machine shown in given figure into Moore machine. **(07)**



OR

A) Minimize the following DFA (If Possible). **(07)**



B) Construct a regular expression for a regular language that generates all the string of 0's and 1's such that each string contains odd 0's and odd 1's.Use Arden's theorem. **(08)**